



Universidad Peruana
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CARRERA DE INGENIERÍA INFORMÁTICA

Curso: SISTEMAS OPERATIVOS

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Kill:

El comando kill finaliza un proceso especificando su numero PID de identificacion individual. Permite enviar senales de cierre normal o forzar la salida inmediata de programas colgados mediante la opcion -9.

```
ubuntu@ubuntu:~$ kill
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
ubuntu@ubuntu:~$ killall
Usage: killall [OPTION]... [--] NAME...
       killall -l, --list
       killall -V, --version

-e,--exact          require exact match for very long names
-I,--ignore-case    case insensitive process name match
-g,--process-group  kill process group instead of process
-y,--younger-than   kill processes younger than TIME
-o,--older-than     kill processes older than TIME
-i,--interactive    ask for confirmation before killing
-l,--list           list all known signal names
-q,--quiet          don't print complaints
-r,--regex          interpret NAME as an extended regular expression
-s,--signal SIGNAL  send this signal instead of SIGTERM
-u,--user USER      kill only process(es) running as USER
-v,--verbose        report if the signal was successfully sent
-V,--version        display version information
-w,--wait           wait for processes to die
-n,--ns PID         match processes that belong to the same namespaces
                    as PID
-Z,--context REGEXP kill only process(es) having context
                    (must precede other arguments)
```

Killall:

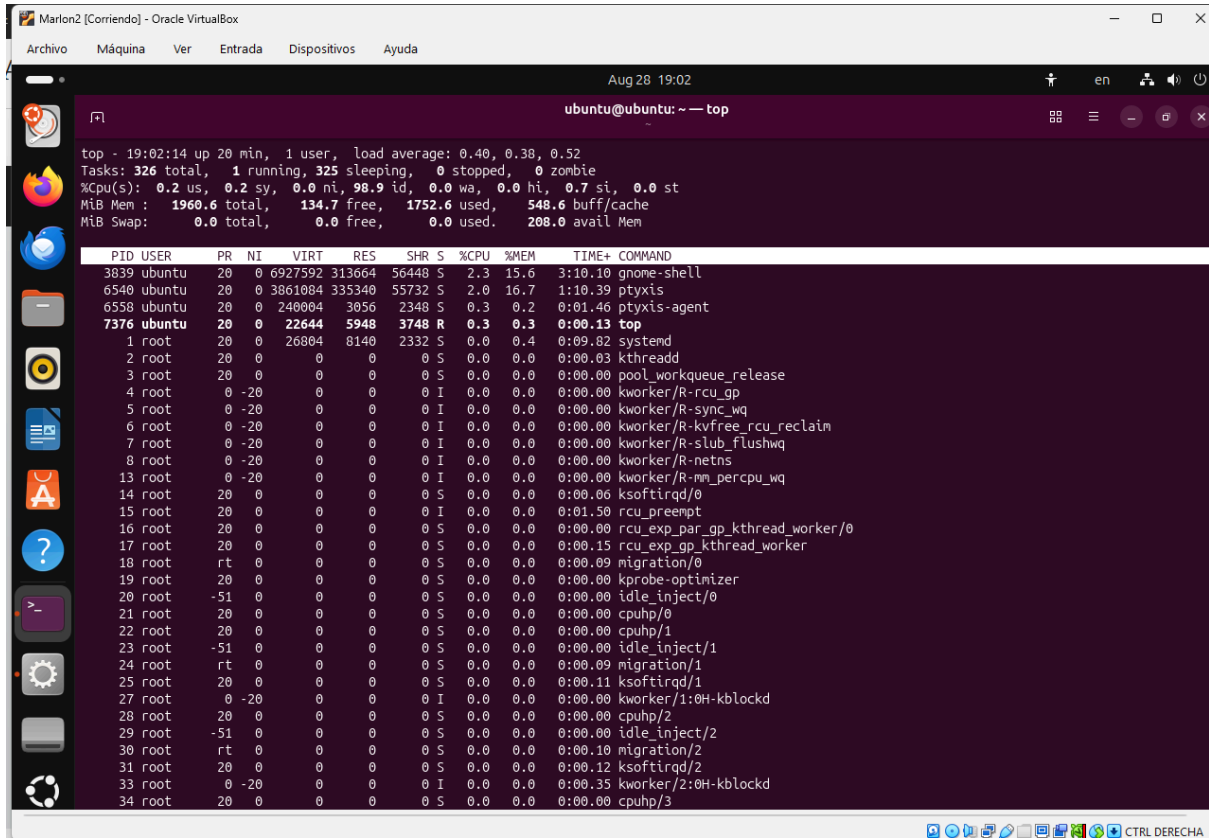
El comando killall detiene procesos buscando directamente por su nombre en lugar de usar un PID. Su ventaja principal es que cierra al mismo tiempo todas las instancias que compartan dicho nombre.

```
ubuntu@ubuntu:~$ killall
Usage: killall [OPTION]... [--] NAME...
       killall -l, --list
       killall -V, --version

-e,--exact          require exact match for very long names
-I,--ignore-case    case insensitive process name match
-g,--process-group  kill process group instead of process
-y,--younger-than   kill processes younger than TIME
-o,--older-than     kill processes older than TIME
-i,--interactive    ask for confirmation before killing
-l,--list           list all known signal names
-q,--quiet          don't print complaints
-r,--regex          interpret NAME as an extended regular expression
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                    as PID
-Z,--context REGEXP kill only process(es) having context
```

Top:

El comando top muestra en tiempo real el uso de recursos del sistema en Linux, como el consumo de procesador, memoria RAM y el estado general de los procesos activos. Funciona como un administrador de tareas continuo donde lista cada programa en ejecucion junto a su identificador PID, usuario, consumo de CPU y memoria.

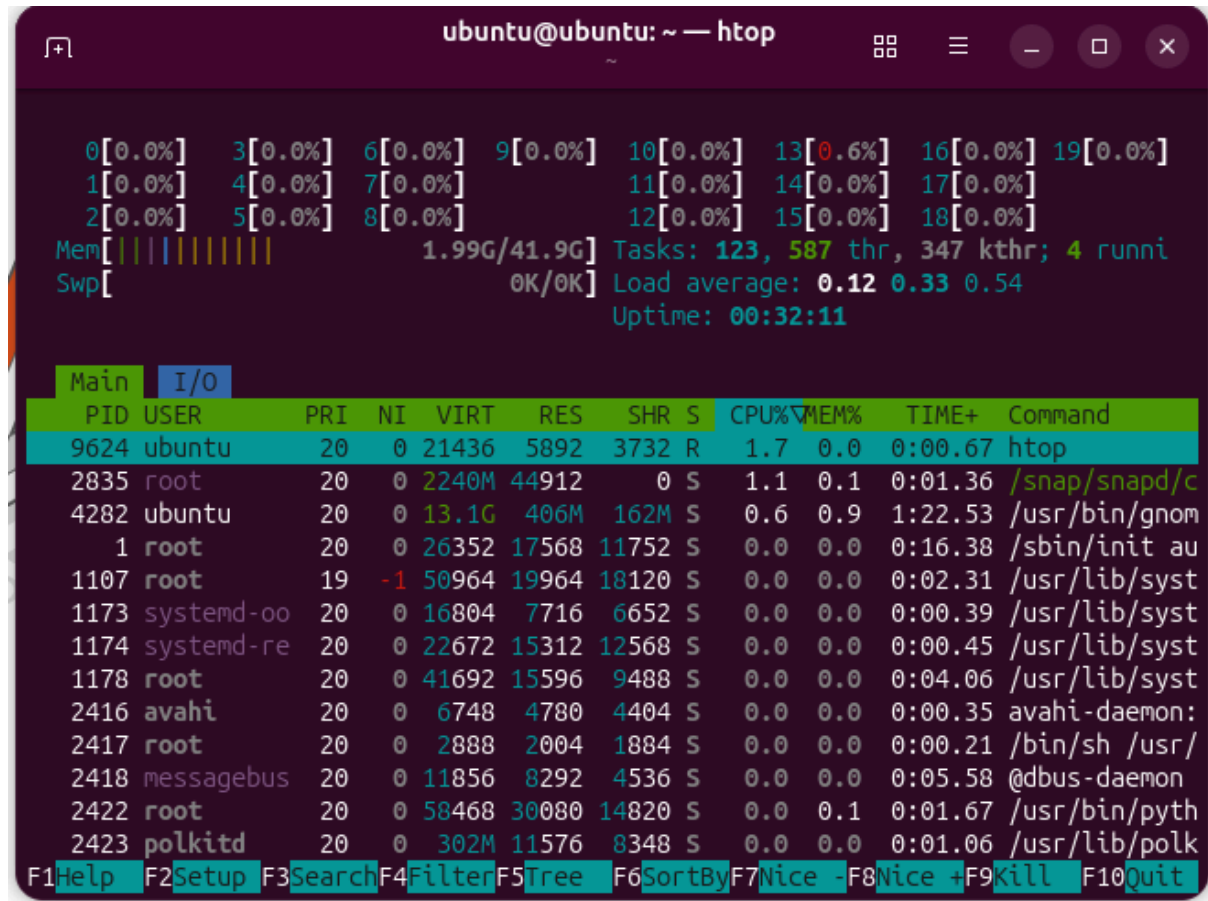


The screenshot shows a terminal window titled 'ubuntu@ubuntu: ~ — top'. The terminal output displays system statistics and a list of running processes. The system statistics include: top - 19:02:14 up 20 min, 1 user, load average: 0.40, 0.38, 0.52; Tasks: 326 total, 1 running, 325 sleeping, 0 stopped, 0 zombie; %Cpu(s): 0.2 us, 0.2 sy, 0.0 ni, 98.9 id, 0.0 wa, 0.0 hi, 0.7 si, 0.0 st; MiB Mem: 1960.6 total, 134.7 free, 1752.6 used, 548.6 buff/cache; MiB Swap: 0.0 total, 0.0 free, 0.0 used, 208.0 avail Mem.

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
3639	ubuntu	20	0	6927592	313664	56448	S	2.3	15.6	3:10.10	gnome-shell
6540	ubuntu	20	0	3861084	335340	55732	S	2.0	16.7	1:10.39	ptxixs
6558	ubuntu	20	0	240004	3056	2348	S	0.3	0.2	0:01.46	ptxixs-agent
7376	ubuntu	20	0	22644	5948	3748	R	0.3	0.3	0:00.13	top
1	root	20	0	26804	8140	2332	S	0.0	0.4	0:09.82	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.03	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0.0	0.0	0:00.06	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:01.50	rcu_preempt
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0.0	0.0	0:00.15	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.09	migration/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kprobe-optimizer
20	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
22	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
23	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
24	root	rt	0	0	0	0	S	0.0	0.0	0:00.09	migration/1
25	root	20	0	0	0	0	S	0.0	0.0	0:00.11	ksoftirqd/1
27	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-kblockd
28	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/2
29	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/2
30	root	rt	0	0	0	0	S	0.0	0.0	0:00.10	migration/2
31	root	20	0	0	0	0	S	0.0	0.0	0:00.12	ksoftirqd/2
33	root	0	-20	0	0	0	I	0.0	0.0	0:00.35	kworker/2:0H-kblockd
34	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/3

Htop:

El comando htop es una versión mejorada e interactiva de top para monitorear los recursos del sistema en Linux. Muestra de forma gráfica el consumo de procesador, memoria RAM y procesos activos, permitiendo desplazarse con el teclado y gestionar tareas fácilmente mediante accesos directos en pantalla.



Nice:

El comando nice permite iniciar un programa en Linux asignándole una prioridad específica de uso de procesador. Si se ejecuta solo, muestra el valor de prioridad actual de la terminal (que por defecto es 0). Un valor mayor significa ser más "amable" y ceder recursos a otros procesos, mientras que un valor menor da más prioridad al programa.

```
daniel-oliva@daniel-oliva-VirtualBox:~$ nice  
0
```